Eccles Building, U.S. Federal Reserve Board of Governors

ECON 100B: Macroeconomics

Class 24: International Trade (Jones Chapter 19)

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April 23, 2026

Agenda

- Announcements
- A core learning objective for **Humanity in 2026**: Smart use of AI
- Last tidbits from Chapter 18:
    - Seignorage: Governments could finance spending by printing money
    - G can crowd out I
- Chapter 19: International Trade
- Next week
    - Chapter 20: Exchange Rates and International Finance
    - In-class Review

Same announcements

- **Problem Set 3** is available, due on **Friday, April 17**
- **Problem Set 4** is available and is due on **Friday, May 1**
- All students get to drop their lowest PS score. If you have scored 100% on PS1, PS2, and PS3, you can skip PS4. But I don't recommend it
- The **Final exam** is confirmed for **Monday, May 11**
  - Just like at midterm, the main class meets here in **Pimentel 1**
  - We will run 90 minutes from about 3:00-4:30PM
  - DSP students: Please either make a proctoring appointment, or for 150% time only, we will have **Anthropology and Art Practice 160**

    (I think it's kind of near Caffe Strada on Bancroft)

Smart use of AI

- Twenty percent of S26 ECON 100B, or about 100 students, incorrectly identified a Nobel laureate on the S26 midterm essay

- I suspect this is because some students use AI suboptimally

- Unless you train an AI chatbot by feeding it background materials, it will answer questions based on <u>its context</u>, not yours

- This is critical to learn now, before you enter the work force

- But let me know if I'm wrong! If you made a different mistake, maybe post it anonymously on Ed Discussion or something

Seignorage = revenue from printing currency

- **Crazy sense:**

  Governments in trouble might just print lots of currency to pay their bills

  See for example Zimbabwe in 2007-09

- **Normal sense:**

  Currency held in your hand literally cannot earn interest

  When the public holds currency, it is effectively extending that government an interest-free loan

  (But most normal governments don't benefit much through this)

Courtesy of Jón Steinsson's ECON 101B

Zimbabwe rolls out Z$100 trillion note

BBC News
January 16th, 2009

Old banknotes such as this, once considered near worthless when in circulation, are enjoying a resurgence in value with tourists eager to buy notes as curiosities. …

The IMF says the [inflation] rate peaked at the end of 2008 at 500 billion percent.

The Scotsman
January 5th, 2010

Government borrowing may **crowd out** investment

- The government borrows money by offering Treasury bills, notes, and bonds to us, the **public**

- By buying a bond, we are “saving” in that we aren’t spending that money

- But we can’t simultaneously save that money and **invest** it in capital

- It’s an either/or kind of thing

    - Either your saving is investment that increases the capital stock

    - Or your saving finances the government’s deficit

    - (Unless you decide to save more and spend less)

- And there’s also the issue of **foreign saving** too ...

How do we see this? Start with income accounting:

$$Y = C + I + G + EX - IM$$

Now rearrange, putting investment, *I*, on one side:

$$I = Y - C - G - EX + IM$$

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$$I = \underbrace{Y - C - T}_{} + \underbrace{T - G}_{} + \underbrace{IM - EX}_{}$$

Income minus consumption minus taxes is what you have left for **private saving**

Taxes minus spending equals **public saving**

Imports minus exports is **foreign saving** (think: imported investment goods)

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Implications:

- If government spending G rises, what happens to investment I? It depends...
- It depends on what happens to private and foreign saving
- Suppose private and foreign saving remain constant
- Then an increase in G without an increase in T will lower public saving, and that reduces investment
- Coming up:
  What happens when foreign saving, a.k.a. the trade deficit IM – EX rises?

Learning objectives for Chapter 19

- International trade has been important for global growth
- Exports represent production and income for the home country
- But imports are not bad. An import becomes C or I, then we subtract with $-IM$ to measure gross domestic product

$$Y = C + I + G + EX - IM$$

- The U.S. has run trade deficits since the 1970s, which have financed high investment and domestic consumption

$$(Y - C - T) + (T - G) - I = EX - IM$$
$$S - I = NX$$

- U.S. trade deficits represent imported foreign saving, and they accumulate to high net foreign ownership of U.S. assets

International trade and Adam Smith’s pin factory

- Like we saw on the S26 midterm essay, trade and specialization are pretty similar

- Within a country, workers and industries within geographic areas specialize in certain activities and then trade with one another

- Activity can cross political borders … and then things get political

- Across all these levels of geography and politics,
    - Trade can be free, and it can also be taxed/tariffed or subsidized
    - Winners are usually numerous, and they mostly all win a little
    - Losers from trade are usually smaller in number, but they lose a lot

In 2026, it's important not to be naïve in any dimension

- Trade
    - Does not magically benefit everyone
    - Can quickly and permanently change people's working lives
    - Is not magically free and open by its nature

- But trade barriers do not magically "re-shore" industries or reverse time in any way

- Trade wars are still wars. If there are winners, there will also be losers who will not appreciate losing

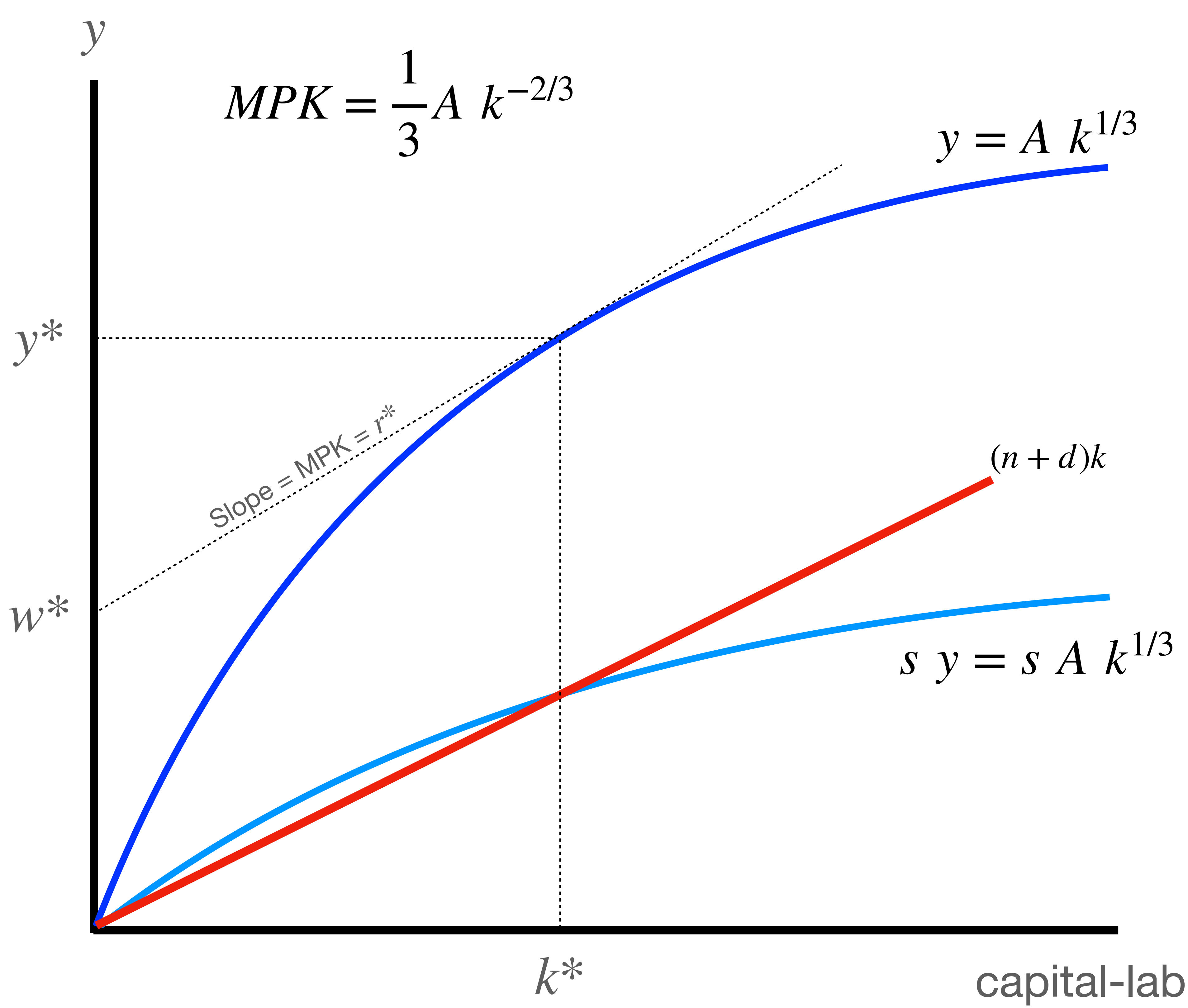
Theory offers some predictions about trade balances, but reality is heterogeneous

- Rapidly growing economies might run **trade deficits** for at least two reasons
  - Because they are growing rapidly, they expect higher incomes in the future and can spend more in the present and borrow against those incomes
  - A trade deficit represents a net inflow of foreign saving, one way of increasing domestic investment that further builds up the domestic capital stock

- But there is also an historical tradition of trying to develop an economy through export-led growth
  - China is a clear example but not the only one. Korea too and other economies in East Asia whose focus was on export markets
  - Kind of a translation of an earlier, seemingly discredited approach called “import substitution” (see Arezki, Brookings 2019)

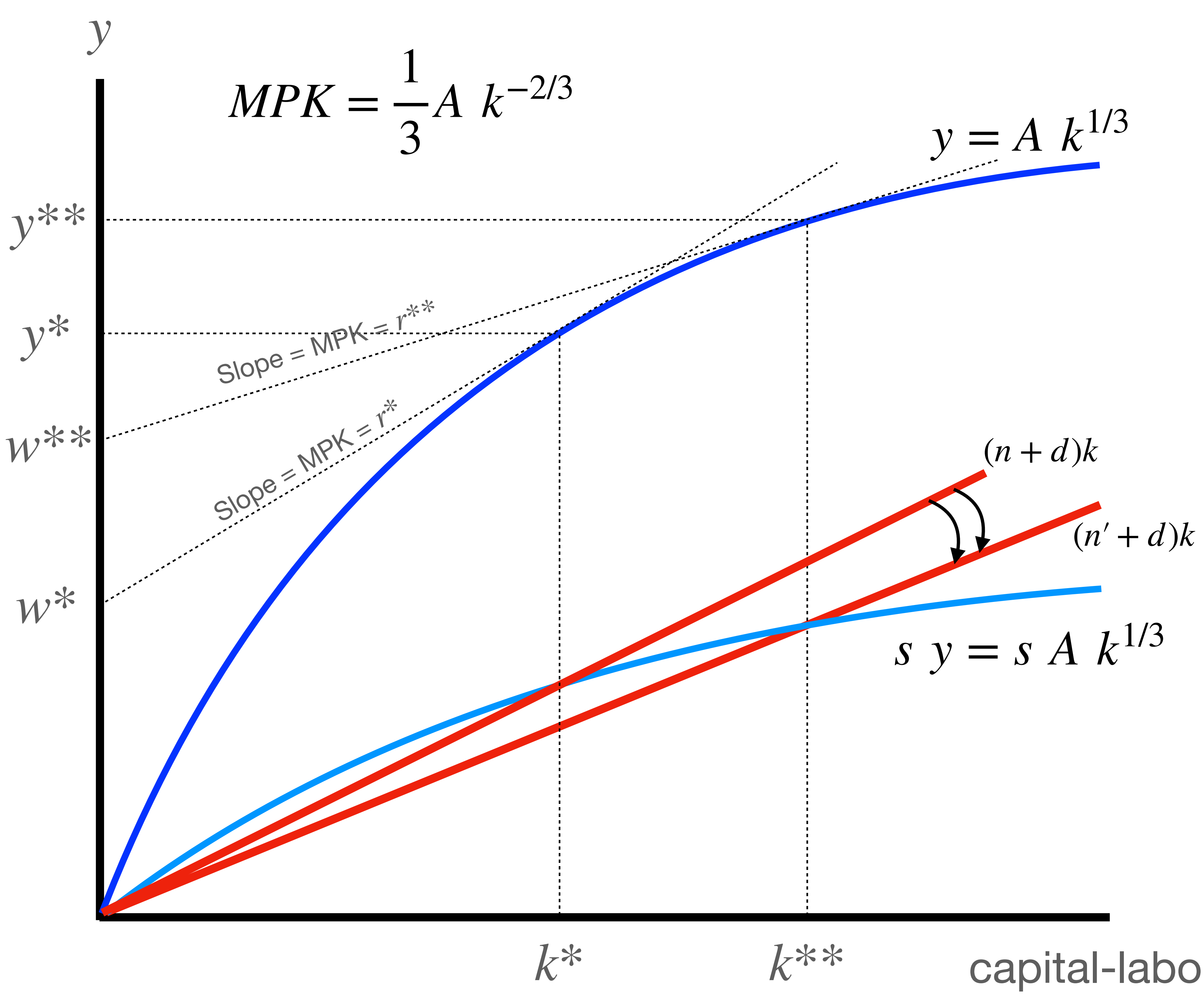
Growth, saving, and capital in the Solow model

Some helpful review of the Solow model



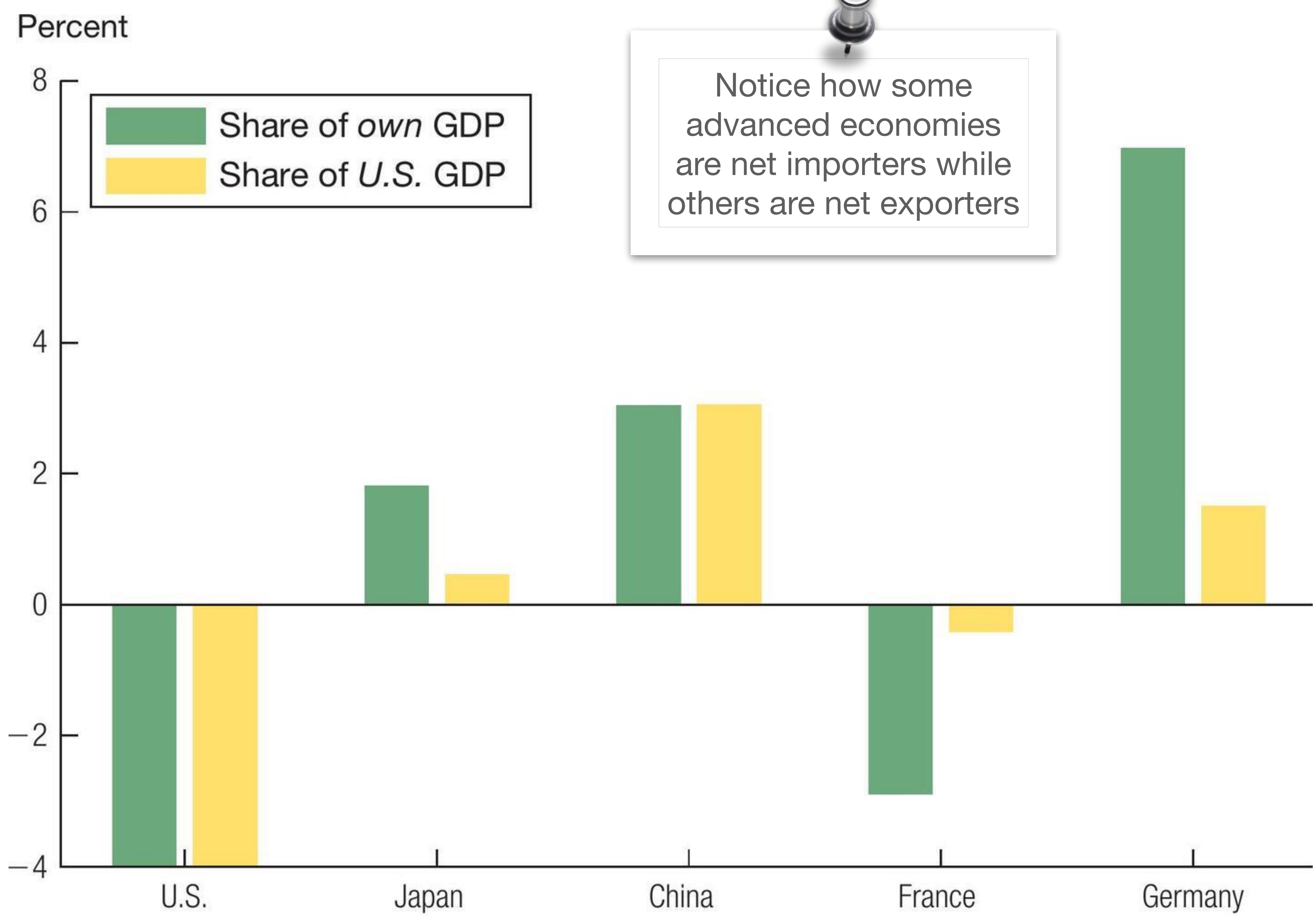
- Hallmarks of human development are:
- Falling mortality
- Then falling fertility
- When fertility rates fall, population growth rates decline
- (And the saving rate might also rise)

A developing country has typically reduced its population growth rate n



- If population growth falls from n to $n' < n$
- Then a Solow economy will endogenously supply the saving required to achieve $k^{**} > k^{*}$
- But importing the saving could be quicker
- In the short run, the marginal product of capital remains high

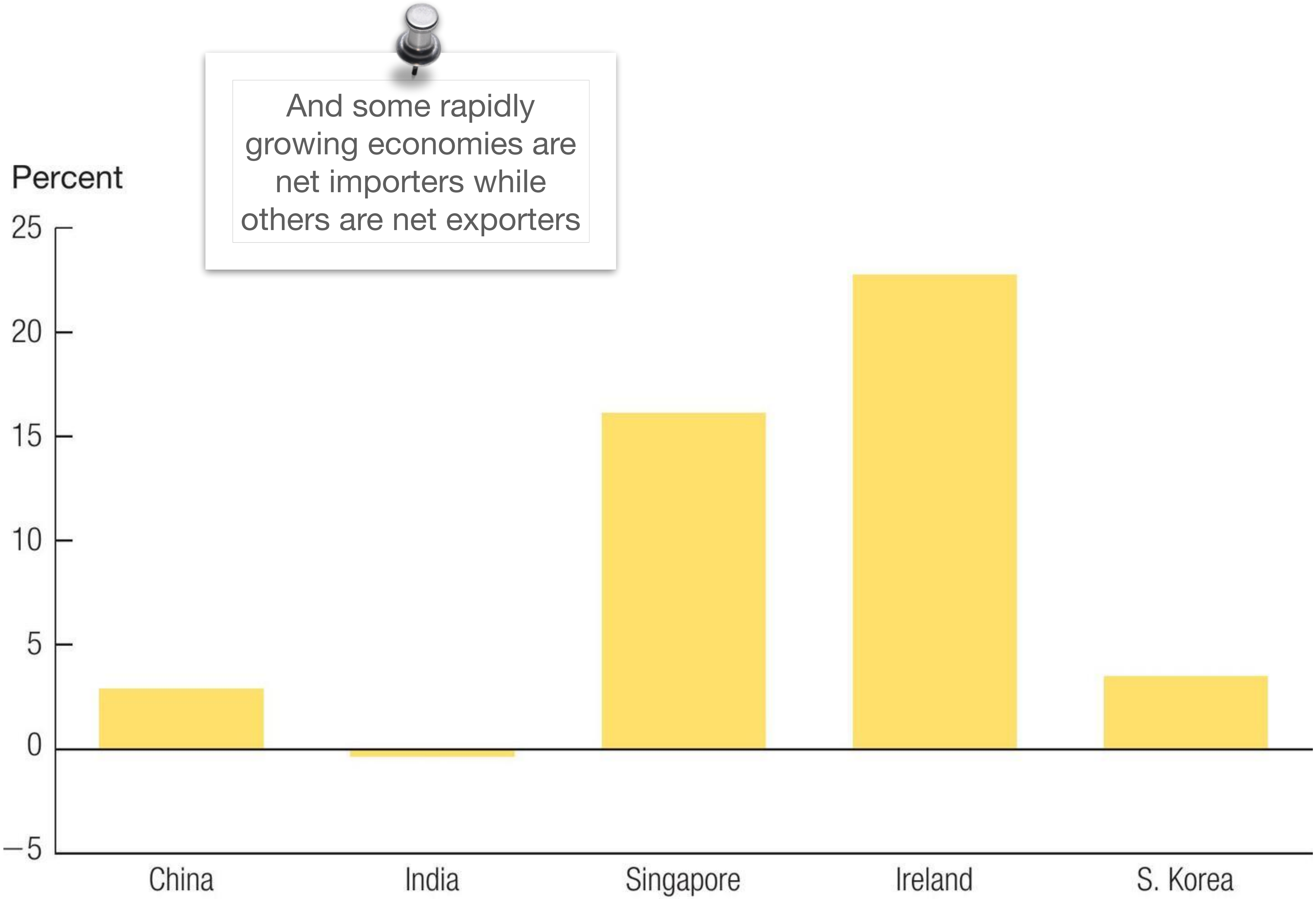
Figure 19.3 The Trade Balance around the World, 2019



Notice how some advanced economies are net importers while others are net exporters

Source: Penn World Table, Version 10.0.

The Trade Balance in Rapidly Growing Countries, 2000–2019



Source: Penn World Table, Version 10.0.

Gains to trade are large, and gains to factor mobility might be even larger

Source: Clemens (JEP, 2011)

Table 1

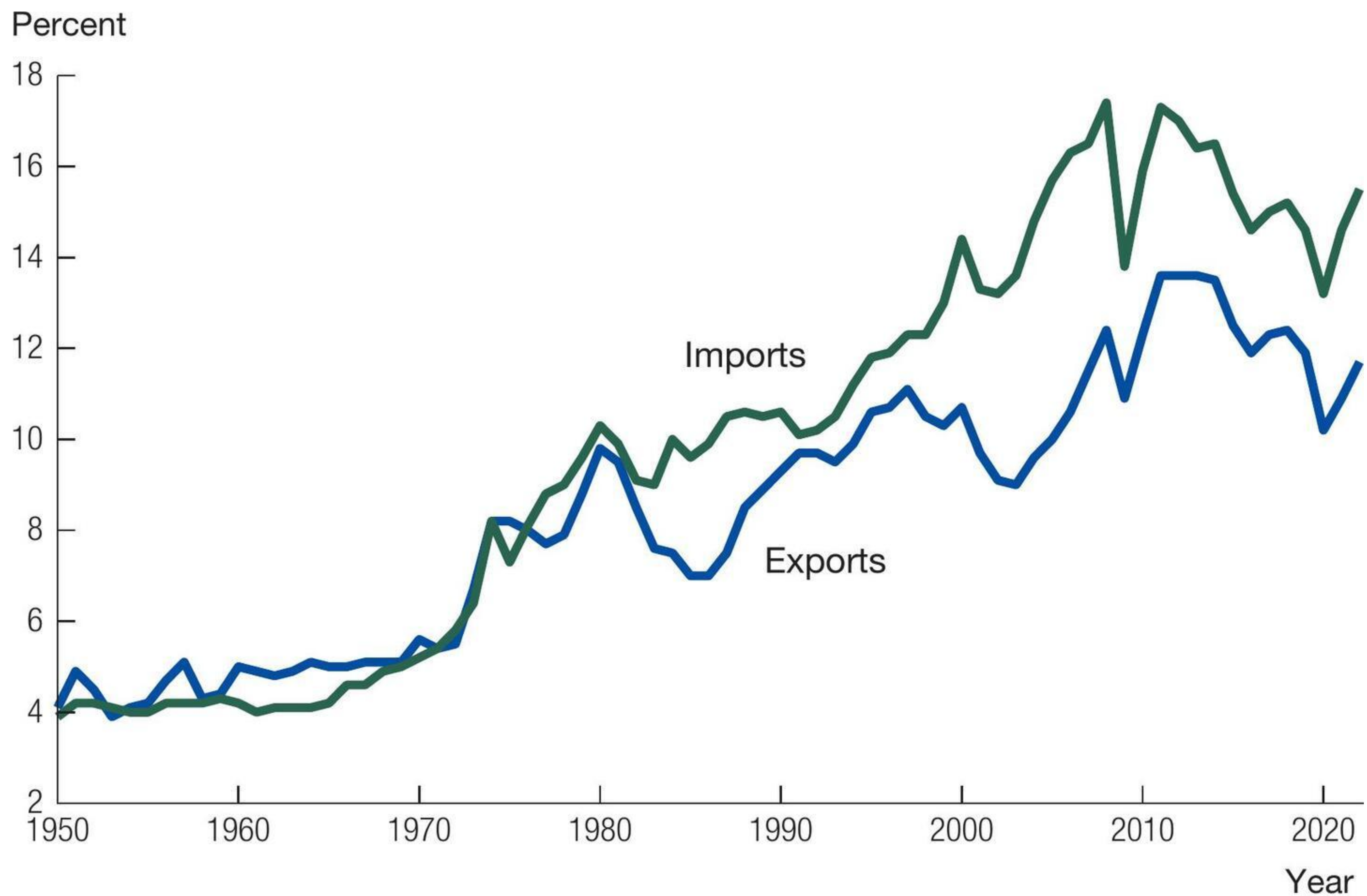
Efficiency Gain from Elimination of International Barriers

(percent of world GDP)

All policy barriers to merchandise trade	
1.8	Goldin, Knudsen, and van der Mensbrugghe (1993)
4.1	Dessus, Fukasaku, and Safadi (1999) [a]
0.9	Anderson, Francois, Hertel, Hoekman, and Martin (2000)
1.2	World Bank (2001)
2.8	World Bank (2001) [a]
0.7	Anderson and Martin (2005)
0.3	Hertel and Keeney (2006, table 2.9)
All barriers to capital flows	
1.7	Gourinchas and Jeanne (2006) [b]
0.1	Caselli and Feyrer (2007)
All barriers to labor mobility	
147.3	Hamilton and Whalley (1984, table 4, row 2) [c]
96.5	Moses and Letnes (2004, table 5, row 4) [c]
67	Iregui (2005, table 10.3) [c,d]
122	Klein and Ventura (2007, table 3) [e]

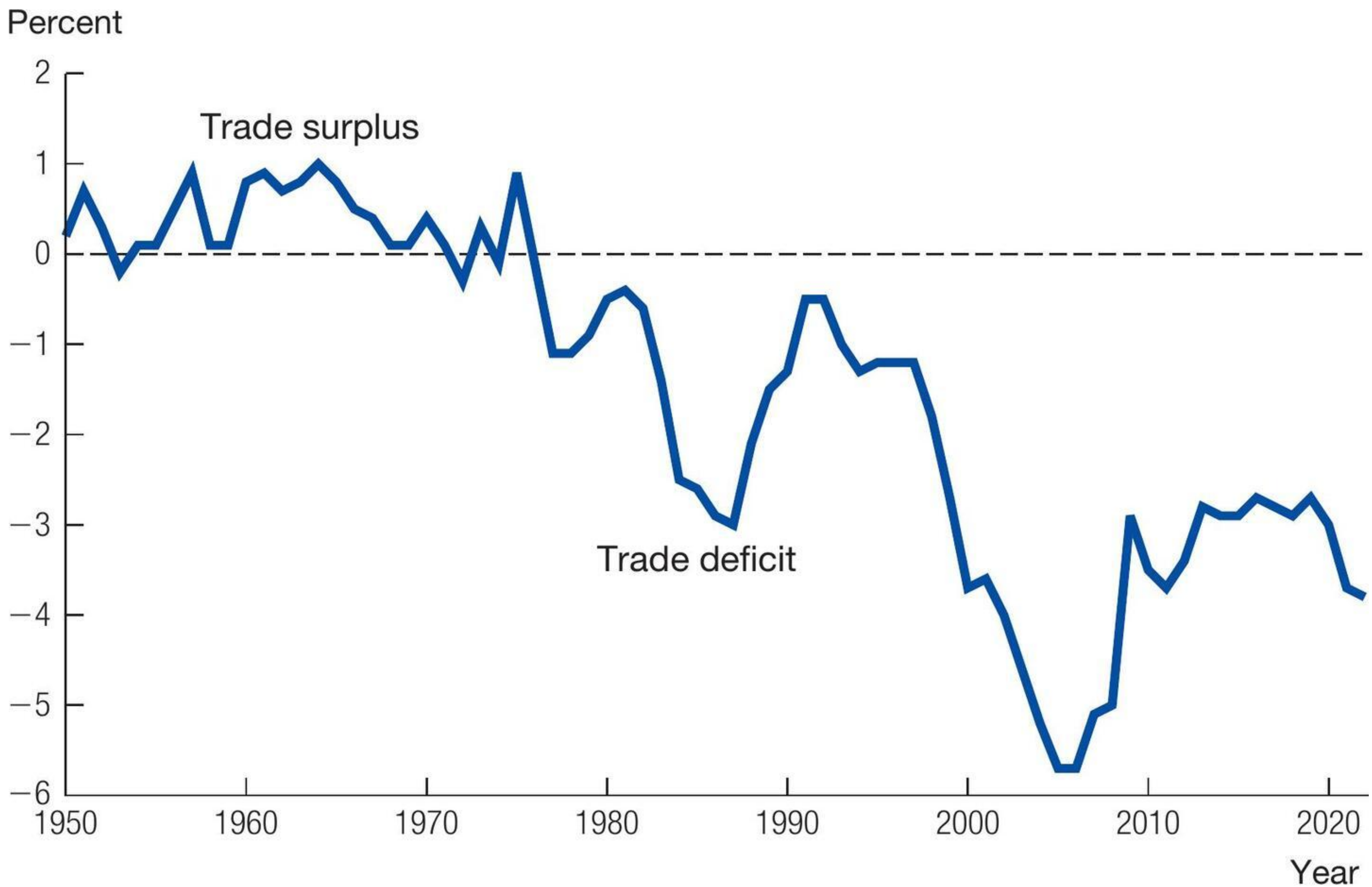
- In addition to trade, another big political issue is immigration, the flow of labor across borders
- Capital can also flow across borders
- Of these three — trade, capital flows, and labor flows — lowering barriers to labor flows would bring the greatest gains

Figure 19.1 U.S. Import and Export Shares of GDP



Source: National Income and Product Accounts.

Figure 19.2 The U.S. Trade Balance as a Share of GDP



Source: National Income and Product Accounts.

Again, income accounting

- This and all these identities derive from $Y = C + I + G + NX$ and then clever arithmetic

- Same math as before, but usually the rearranging ends up:

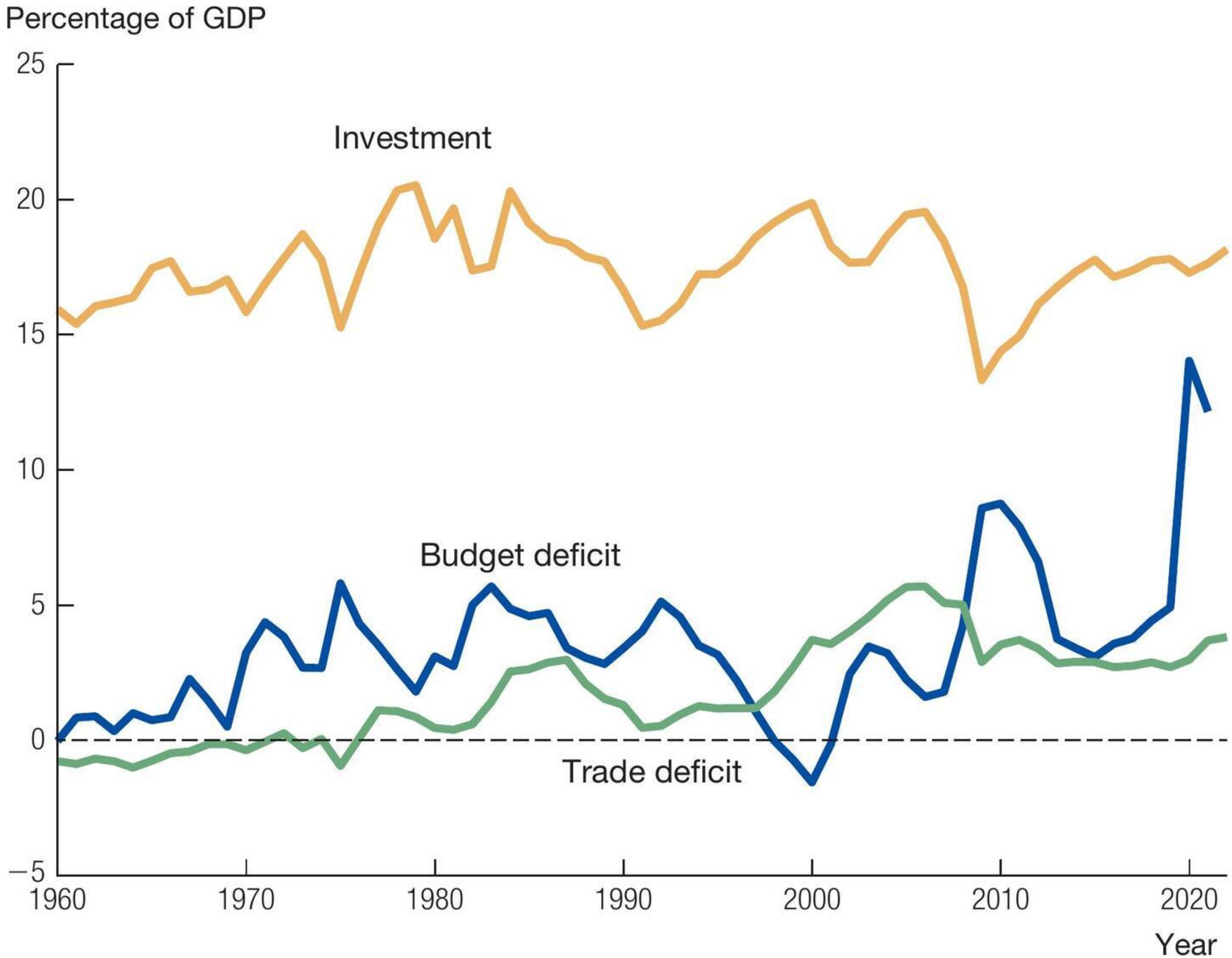
$$S - I = NX$$

  domestic saving minus domestic investment = trade balance

- What is challenging: Each element S, I, NX can be moving around

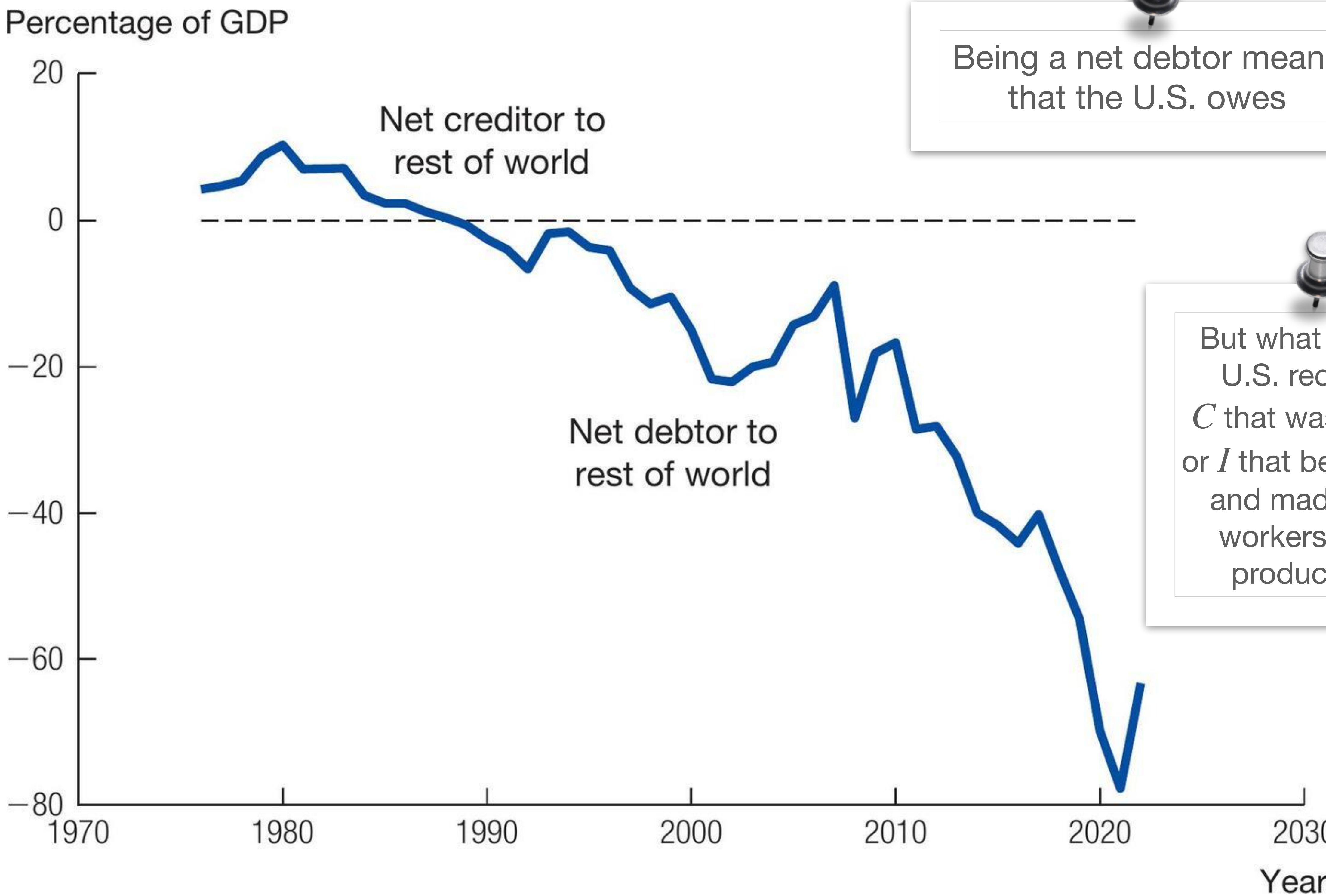
- What is helpful: The U.S. story that emerges is one of **strong domestic investment** and low domestic saving, especially public saving. A trade deficit must be the result

Figure 19.5 U.S. Investment, Budget Deficit, and Trade Deficit



Source: National Income and Product Accounts.

Figure 19.6 Net International Investment Position of the U.S.



Being a net debtor means that the U.S. owes

But what did the U.S. receive? C that was spent, or I that became K and made U.S. workers more productive?

Source: U.S. Bureau of Economic Analysis, U.S. Net International Investment Position [IIPUSNETIQ], retrieved from FRED, Federal Reserve Bank of St. Louis; fred.stlouisfed.org/series/IIPUSNETIQ.

Takeaways

- Trade deficits are not necessarily signs of weakness or unfair trade practices
- Importing foreign saving can fuel strong investment
- But lots of government borrowing — dissaving — might be something the U.S. could change, and that would probably reduce the trade deficit